PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Bioinputs as a Sustainable Strategy for Rice and Bean Production: Implications for Human Health and Planetary Nutrition	
ABSTRACT			
Abstract	2	The abstract has not yet been structured.	
INTRODUCTION			
Rationale	3	This review is grounded in the growing demand for sustainable food systems capable of reducing the environmental impacts of conventional agriculture, increasing the resilience of production systems, and ensuring continuous access to healthy foods in the face of climate change and population growth, thereby promoting healthy dietary patterns. It establishes a relationship between the use of bioinputs and the production of rice (Oryza sativa) and common bean (Phaseolus vulgaris), staple crops for Brazilian food security. The analysis is conducted in light of the principles of planetary nutrition, emphasizing the interrelationships among agricultural sustainability, human health, and environmental conservation.	
Objectives	4	**General objective:** To critically analyze the available scientific evidence on the application of bioinputs in rice (Oryza sativa) and common bean (Phaseolus vulgaris), as well as their relationships with agricultural sustainability and the promotion of planetary health. **Specific objectives:** Identify the types of bioinputs used in rice and bean production, highlighting their functions (biofertilizers, biocontrol products, biostimulants, etc.); Assess the agronomic impacts of bioinput use on productivity and soil health in these crops; Investigate the contributions of bioinput use to reducing dependence on synthetic inputs and the associated environmental benefits; Analyze how the adoption of bioinputs relates to the principles of planetary nutrition, considering aspects of human health and the sustainability of food systems.	
METHODS			
Eligibility criteria	5	Peer-reviewed scientific studies addressing the use of bioinputs in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production will be included, considering the categories of biofertilizers, inoculants, biostimulants, performance enhancers, and biocontrol products. For this review, bioinputs will be understood in accordance with Brazilian Law No. 15,070 of December 23, 2024, as products, processes, or technologies of biological origin (plant, animal, or microbial), including biotechnology-based products, intended for agricultural production, protection, storage, and processing, that interfere with growth, organism response mechanisms, and interact with physicochemical and biological processes. Studies using any form of application, including foliar application, soil application, seed treatment, or other management strategies, will be accepted, provided that bioinput use is clearly described and linked to a measurable assessment of agronomic effects. There will be no restriction regarding crop phenological stage, provided that the effects are evaluated clearly, objectively, and measurably. Publications in Portuguese, English, and Spanish will be considered, with no restriction regarding the climate of origin of the study, provided that the results are applicable to the context of Brazilian agriculture. Experimental studies, systematic reviews, and meta-analyses will be accepted, provided that they meet methodological criteria of scientific quality, including the absence of undeclared conflicts of interest, clear definition of the type of publication, and sufficient methodological robustness to support the conclusions presented. Duplicate studies, publications not peer reviewed, reports without a defined methodology, opinion articles, letters to the editor, and publications without practical application related to the agricultural management of rice and common bean will be excluded. In addition, a time frame covering publications indexed from 2015 to 2025 will be adopted, ensuring the currency of the evidence used in this review.	
Information sources	6	ScienceDirect - time / research articles PubMed - time / free full text Web of Science - time / article SciELO - apply the filters after the search and verify Frontiers in Science - Original Research / Past year	

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		(Databases with little return will be removed from the projects; tests should be performed, as, for example, PubMed does not understand the filters used for selection).	
Search strategy	7	**First attempt:** ("bioinputs" OR "biofertilizers" OR "biocontrol products" OR "biostimulants" OR "bioinoculants" OR "plant extracts" OR "botanical pesticides" OR "microbial fertilizers" OR "algae-based inputs") AND ("Oryza sativa" OR "rice" OR "Phaseolus vulgaris" OR "common bean" OR "arroz" OR "feijão") AND ("sustainable agriculture" OR "agroecology" OR "planetary health" OR "sustainability" OR "planetary nutrition") If no results are returned, test the searches below one by one. **Second attempt:** ("bioinputs" OR "biofertilizers") AND ("Oryza sativa" OR "rice") ("bioinputs" OR "biofertilizers") AND ("Phaseolus vulgaris" OR "common bean") ("bioinputs" OR "biofertilizers") AND ("arroz" OR "feijão") ("biocontrol products" OR "biostimulants") AND ("Oryza sativa" OR "rice") ("biocontrol products" OR "biostimulants") AND ("Phaseolus vulgaris" OR "common bean") ("algae-based inputs" OR "plant extracts") AND ("Oryza sativa" OR "rice") ("algae-based inputs" OR "plant extracts") AND ("Phaseolus vulgaris" OR "common bean") ("algae-based inputs" OR "plant extracts") AND ("arroz" OR "feijão") ("algae-based inputs" OR "plant extracts") AND ("Oryza sativa" OR "rice") ("algae-based inputs" OR "plant extracts") AND ("Phaseolus vulgaris" OR "common bean") ("algae-based inputs" OR "plant extracts") AND ("arroz" OR "feijão")	
Selection process	8	The selection of studies will be conducted by three independent reviewers, including a specialist in bioinputs and rice and bean management. Initially, a search will be performed using predefined descriptors selected to align with the overall scope of the study. Subsequently, titles and abstracts will be screened to identify studies addressing the main topic. Only studies that fully meet the predefined inclusion criteria will be selected for full-text review. These include studies evaluating the use of bioinputs applied to the management of rice and/or common bean, published in peer-reviewed journals, with full-text available in English, Portuguese, or Spanish, published within the last ten years, and presenting original results with empirical data directly related to the proposed topic. Opinion papers, narrative reviews, editorials, letters to the editor, conference abstracts, duplicate studies, or studies without full-text access will be excluded, as well as those not primarily focused on bioinputs or addressing crops other than rice and common bean. The entire screening process will be conducted independently and in a blinded manner by the reviewers using the Rayyan platform, following the PRISMA flow diagram. After full-text assessment, eligible studies will be included for data extraction and analysis. The process will be conducted in duplicate, ensuring agreement among reviewers and resolving any discrepancies through consensus during technical meetings. The methodological quality of the included studies will be assessed based on predefined criteria, including the clarity of objectives and hypotheses, the adequacy and detail of the experimental design, the consistency and validity of statistical methods, the completeness and transparency of reported results, and the assessment of risk of bias. This step aims to ensure the robustness of the evidence and the reliability of the conclusions. The review will be registered in PROSPERO, and following the final selection, systematic data extraction will be carried out according to the variables of interest defined in the protocol.	
Data collection process	9	Information will be collected regarding the type and category of bioinput (e.g., biofertilizer, biopesticide, or biostimulant), as well as the commercial or scientific name of the bioinput when available, the author and year of publication, the target crop (rice, common bean, or both), the application method (e.g., soil, foliar, seed treatment), the applied dose, the timing or phenological stage of application, the frequency of use, and the field history. Field history will be considered only when clearly described and including previous management practices, prior crops, input use, and soil conditions.	

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		In addition to these variables, key agronomic outcomes will be extracted, including crop productivity, plant health, nutrient use efficiency, and pest and disease control. Environmental impacts will also be recorded, including effects on soil, biodiversity, reduction in agrochemical use, and potential implications for human health or food security. Contextual data will also be collected, such as edaphoclimatic conditions, study location, experimental design, and information on funding sources and authors' institutional affiliations.	
Data items	10a	Data will be collected on rice and common bean productivity, resource use efficiency, biological control of pests and diseases, environmental impact, and soil health.	
	10b	Information will be collected regarding the type of study, geographic region, edaphoclimatic conditions, funding sources, and authors' institutional affiliations.	
Risk of bias assessment	11	The risk of bias of the included studies will be assessed using the Joanna Briggs Institute (JBI) critical appraisal tools, adopting specific versions for cross-sectional and intervention studies. Three reviewers will independently and blindly apply the instruments to ensure reliability and minimize potential subjective influences during the assessment. For cross-sectional studies, criteria such as the clarity of inclusion criteria, detailed description of participants and setting, validity of measurement instruments, adequacy of statistical methods, and identification and control of confounding factors will be evaluated. For intervention studies, aspects including the clarity of group allocation, similarity of groups at baseline, reliability of outcome measurement methods, appropriate follow-up procedures, and data analysis based on the intention-to-treat principle (when applicable) will be considered. Additionally, compliance-related risk of bias will be assessed, considering potential conflicts of interest, declared funding sources, authors' institutional affiliations, and any indication that these factors may have influenced the study design, conduct, or interpretation of results. Each criterion will be rated as "yes", "no", "unclear", or "not applicable", according to the tool's guidelines. Discrepancies between reviewers will be resolved by consensus or, when necessary, by consultation with a fourth reviewer. The results of the risk of bias assessment will be used not only as a methodological quality filter but also to support the critical interpretation of findings and to inform the overall strength of evidence derived from each study.	
Effect measures	12	This review is primarily qualitative, focusing on the critical and descriptive analysis of the results reported in the selected studies. However, in cases where data are comparable across studies, particularly regarding variables such as crop productivity, bioinputs, sustainability indicators, or environmental metrics—a complementary quantitative analysis will be conducted.	
Synthesis methods	13a	Studies will be organized thematically according to: crop type (rice or common bean), type and class of bioinput (e.g., biofertilizers, biopesticides, plant growth promoters, plant extracts, microorganisms), agronomic impact, environmental implications, and potential effects on human health.	
	13b	A segmentation will be performed by crop (rice or common bean) and, within each group, by the agronomic purpose of the bioinput: yield improvement, pest and disease control, soil health enhancement, and resource use efficiency.	
	13c	Bioinputs will be classified according to their origin (microbial, plant-based, mineral, or combined), mode of action (e.g., nutritional, protective, symbiotic), and application method (soil, foliar, or seed treatment). This classification will enable the comparison of different technological approaches.	
	13d	The potential of bioinputs to promote sustainable practices will be assessed, with emphasis on environmental impacts (e.g., reduction in pesticide and chemical fertilizer use, soil health, and efficient water use) and potential implications for human health, both directly and indirectly.	
	13e	The narrative synthesis will also examine the consistency of findings across studies, identifying recurring patterns, discrepancies, and potential sources of heterogeneity, such as experimental design, climate, soil conditions, or socioeconomic context.	

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13f	Finally, a critical analysis of the gaps identified in the literature will be conducted, particularly regarding the lack of long-term studies, integrated assessment of impacts, and data from key regions for food security. These gaps will inform recommendations for future research.	
Assessment of reporting bias	14	In addition, transparency in methodological reporting will be assessed, particularly regarding experimental design, control of variables, and funding sources. Studies with a high risk of bias (e.g., absence of a control group, incomplete reporting of results, or undeclared conflicts of interest) will be identified, and the biases will be discussed within the review. This assessment will seek to understand the extent to which the available data reflect a realistic and balanced view of the effects of bioinputs on rice and common bean, considering their implications for agricultural sustainability and planetary health.	
Assessment of certainty	15	The assessment of the overall confidence in the results will be conducted through a critical appraisal of the methodological quality of the included studies. Aspects such as the clarity of objectives, the adequacy of experimental designs, the description of statistical methods, the consistency between results and conclusions, and the potential generalizability of the findings will be considered. In addition, the possibility of publication bias will be evaluated by examining the predominance of positive findings regarding the effects of bioinputs, the absence of studies from certain geographic regions—particularly those with significant relevance for rice and common bean production, and the recurrence of publications originating from a limited number of institutions or research groups. These factors may indicate a concentration of evidence and limit the diversity of evaluated contexts.	
RESULTS			
Study selection	16a	The selection of studies will be conducted in a standardized, supervised, and documented manner to ensure reproducibility and consistency of the process. Initially, all records identified in the databases will undergo screening by title and abstract based on predefined inclusion criteria: (i) explicit use of bioinputs in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production; (ii) availability of data applicable to agricultural contexts; (iii) publication in peer-reviewed journals; (iv) language in English, Portuguese, or Spanish; and (v) alignment with themes related to agricultural sustainability and/or planetary nutrition. At this stage, duplicate studies, those outside the thematic scope, or those with insufficient information for preliminary assessment will be excluded. The selected articles will proceed to full-text review, during which the same inclusion criteria will be applied more rigorously. Each study will be independently assessed by three reviewers and classified according to an eligibility score: 0 = not eligible; 1 = eligible with reservations; 2 = fully eligible. The score will be assigned based on the study's adherence to the inclusion criteria and the clarity of its methodological description. Studies with discrepant scores among reviewers will be discussed in a joint technical meeting until consensus is reached. If consensus cannot be achieved, a fourth reviewer will be consulted.	
	16b	The initial search strategy will identify [n] records. After the removal of duplicates and screening based on the established criteria, [n] studies will be included in the review. A PRISMA flow diagram will be presented to detail the study selection process.	
Study characteristics	17	The final selection of studies will be based on the presence of key characteristics that ensure their relevance and alignment with the scope of the review. Only studies that clearly and objectively report the application of bioinputs—such as biofertilizers, microbial inoculants, biopesticides, biostimulants, or other biologically based products—in rice (Oryza sativa) and/or common bean (Phaseolus vulgaris) production systems will be included. It is essential that the studies describe the application method, the agronomic context in which they were conducted (including edaphoclimatic conditions, management practices, and crop phenological stage), and the observed effects on agronomic, environmental, or productivity-related parameters.	

PRISMA 2020 Checklist

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Risk of bias in studies	18	The risk of bias of the included studies will be described based on the results of the assessment conducted using the Joanna Briggs Institute (JBI) tool, as outlined in item 11 of this protocol. Studies will be classified according to their level of risk (low, moderate, or high) based on the criteria established by the tool, considering methodological aspects such as the clarity of inclusion criteria, validity of measurement instruments, control of confounding factors, and the presence of an appropriate control group, among others. The frequency and distribution of risk of bias levels across studies will be presented in a dedicated table, and these data will be considered in the analysis of the reliability of the evidence during the synthesis of results.	
Results of individual studies	19	Relevant data from the included studies will be organized in a standardized data extraction table, presenting the key information of each article, including: year of publication, country where the study was conducted, study design, evaluated crop (rice and/or common bean), type and category of bioinput used, application method, and the main reported agronomic outcomes (such as productivity, plant health, soil indicators, or input use). When available, information on funding sources and institutional affiliations will also be recorded. These data will be used to support the descriptive synthesis of the review findings.	
Results of syntheses	20a	The included studies will be grouped into thematic syntheses according to the crop (rice or common bean), the type and class of bioinput (e.g., biofertilizers, biopesticides, plant growth promoters, plant extracts, among others), the reported impacts (such as productivity, pest and disease control, and soil health), and their implications for agricultural sustainability and planetary health.	
	20b	The essential characteristics of the studies included in the review will be organized in a standardized data extraction table, developed based on the criteria previously defined during the selection stage. This table will aim to clearly and objectively summarize the most relevant information from each study, ensuring consistency with the methodological criteria described earlier. This systematization is intended not only to facilitate comparative analysis across studies but also to identify response patterns, thematic gaps, and potential institutional or regional influences on the reported results.	
	20c	The main findings of each study will be described individually and in a summarized manner, highlighting the observed effects of bioinputs in rice or common bean production systems. Data such as yield increases, effectiveness in pest and disease control, improvements in soil health, and reductions in the use of chemical inputs will be reported when available.	
	20d	The narrative synthesis of the results will highlight the positive effects attributed to the use of bioinputs in rice and common bean production systems, such as increased productivity and reduced environmental impacts. Variability in findings across studies will be discussed, considering factors such as the type of bioinput, experimental conditions, and application context. Identified gaps will include the scarcity of long-term studies and the limited availability of data on impacts related to human health and the environment at a regional scale, which may restrict broader inferences regarding the role of bioinputs as a strategy for planetary nutrition.	
Reporting biases	21	The possibility of publication bias will be considered qualitatively and critically discussed in the synthesis of results. For this review, positive results will be defined as studies reporting beneficial effects attributed to the use of bioinputs, such as increased productivity, improved soil health, effective pest or disease control, or significant reductions in the use of synthetic inputs, compared to control or conventional treatments. In experimental studies involving bioinputs, the main risks of bias include: (i) absence of an appropriate control group or use of non-equivalent controls; (ii) lack of blinding during the conduct of the experiment and outcome assessment; (iii) small sample sizes, which may compromise statistical robustness; (iv) omission of negative or neutral results, potentially inflating the perceived effectiveness of bioinputs; (v) undisclosed conflicts of interest, particularly in studies funded by industry; and (vi) incomplete reporting of management conditions, environmental context, or product composition.	

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
		Additionally, the geographic concentration of publications, the recurrence of specific authors or institutions, and funding patterns will be examined, as these may indicate institutional bias. Methodological transparency in data reporting will also be assessed, including the disclosure of variability, limitations, and applied statistical tests. These elements will be integrated into the final discussion of the review to provide a critical and contextualized interpretation of the level of confidence in the available evidence regarding the effects of bioinputs in rice and common bean production systems.	
Level of significance	22		
DISCUSSION			
Discussion	23a	Bioinputs demonstrate significant potential to promote the transition from conventional agricultural systems to more sustainable food systems, particularly in staple crops in Brazil, such as rice and common bean. The results indicate that the use of biofertilizers, biopesticides, and plant growth promoters can contribute to increased productivity, reduced reliance on synthetic inputs, and improved soil health, key elements of planetary nutrition and agricultural sustainability.	
	23b		
	23c		
	23d		
OTHER INFORMATION			
Protocol registration	24a	The review has not yet been registered in a public platform such as PROSPERO, but it will be registered once the protocol is finalized and approved, in accordance with PRISMA guidelines. The registration link will be included in the manuscript at a later stage.	
	24b		
	24c		
Funding	25	No funding.	
Conflicts of interest	26	The authors declare no conflicts of interest.	
Availability of data, code, and other materials	27	The extracted data, data collection forms, and analytical materials will be available upon request from the corresponding author.	

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